Tina Radvand

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Education

Ph.D. in Civil Engineering (Transportation Engineering) Jan 2022 – Present
University of Illinois Urbana-Champaign
Advisors: Prof. Alireza Talebpour and Prof. Yanfeng Ouyang

Master of Science in Civil Engineering May 2025
University of Illinois Urbana-Champaign
Advisor: Prof. Alireza Talebpour

B.Sc. in Civil and Environmental Engineering Sep 2016 – Jan 2021
Sharif University of Technology

Research Interests

Network Optimization, Quantum Computing, Reliability, Transportation

Publications

- T. Radvand, Y. Ouyang, and A. Talebpour. *Quantum Approximate Optimization for Facility Location Problems*. Manuscript in preparation, expected submission 2026.
- T. Radvand, Y. Ouyang, and A. Talebpour. *Minimum Path Decomposition in Directed Acyclic Graphs*. In preparation, expected submission 2026.
- T. Radvand, Y. Ouyang and A. Talebpour. *Reliable Facility Location Design under Correlated Disruptions and Imperfect Information.*. In preparation, expected submission 2026.
- S. Chung, T. Radvand, and A. Talebpour. *Evolution of Lane-Changing Behavior in Mixed Traffic: A Quantum Game Theory Approach*. Under review at *IEEE Transactions on Intelligent Transportation Systems*, available on ArXiv 2026.
- T. Radvand, A. Talebpour, and Y. Ouyang. *Hierarchical Pod Routing for Modular Transit Systems*. Submitted to Transportation Research Part C: Emerging Technologies, 2026. Available on arXiv.
- T.Radvand, Y. Zhang, A. Talebpour and S. Hamdar. *Dynamics of Traffic Jam Formation and Dissipation.* In preparation, expected submission 2026.
- T. Radvand, A. Talebpour and H. Khosravian. *A Quantum Optimization Algorithm for Optimal Electric Vehicle Charging Station Placement for Intercity Trips.* Under review at *IEEE Transactions on Intelligent Transportation Systems*, available on ArXiv, 2024.
- R. Ammourah, et al. *Introduction to the Third Generation Simulation Dataset: Data Collection and Trajectory Extraction.* Transportation Research Record, 2024.
- A. Talebpour, Y. Zhang, T. Radvand, M. Yousefi *Advancing Self-Enforcing Streets Phase 1: The Relationship between Roadway Environment and Crash Severity, Illinois.* Department of Transportation, 2024.
- T. Radvand, H. Khosravian and A. Talebpour. *A Quantum Annealing Approach to Discretized Traffic Assignment Problem.* Presented at the TRB 103rd Annual Meeting, January 2023.

Presentations

- *Entangled Traffic: A Quantum Game-Theoretic Framework for Lane-Changing Cooperation in Mixed Traffic*
  hEART, Paris, France, September 2026.
- *Reliable Facility Location Design under Correlated Disruptions and Imperfect Information.*
  TRB Annual Meeting, Washington, D.C., January 2027;
  INFORMS Annual Meeting, San Francisco, CA, November 2026;
  Transportation Science and Logistics Conference, Cambridge, MA, July 2026.
- *Rebalancing Modular Transit Systems: Graph-Based Optimization for Fleet Size Planning and Routing.*

INFORMS Annual Meeting, Atlanta, GA, October 2025;
TRB Annual Meeting, Washington, D.C., January 2026.

- *A Quantum Optimization Algorithm for Electric Vehicle Charging Station Placement for Intercity Trips.*
  TRB Annual Meeting, Washington, D.C., January 2025;
  Global Symposium on Mobility Innovation (CCAT), Ann Arbor, MI, May 2024;
  INFORMS Annual Meeting, Seattle, WA, October 2024;
  INFORMS Annual Meeting, Phoenix, AZ, October 2023.
- *Coexistence in Motion: Unveiling the Behavioral Dynamics of Humans and Highly Automated Vehicles in Naturalistic Mixed Traffic.*
  TRB Annual Meeting, Washington, D.C., January 2025.
- *Dynamics of Traffic Jam Formation and Dissipation.*
  TRB Annual Meeting, Washington, D.C., January 2025.
- *Man vs. Machine: A Closer Look at the Differences Between Human Driver Behavior and Automated Vehicle Decision-Making.*
  TRB Annual Meeting, Washington, D.C., January 2024.
- *A Quantum Annealing Approach to Discretized Traffic Assignment Problem.* TRB Annual Meeting, Washington, D.C., January 2023.
- *A Quantum Computing Approach to Discretized Traffic Assignment Problem.* INFORMS Annual Meeting, Indianapolis, IN, October 2022.
- *Unmanned Aerial Vehicle Routing: A Quantum Computing Approach.* TSL Conference, Chicago, IL, 2023.

Awards and Honors

Graduate Student Representative, Institute of Transportation Engineers at UIUC, Student Chapter	2024
Member of Chi Epsilon, National Civil Engineering Honor Society, USA	2024
Recognized as Special Talented Student, Sharif University of Technology	2020
Educational Award, Second Student Recognition Ceremony, Civil Engineering Department, Sharif University of Technology	2017
National Silver Medal, 11th Iranian Astronomy and Astrophysics Olympiad, Iran	2015
Member of National Elite Foundation, Iran	2015

Teaching Experience

Teaching Assistant, Soil Mechanics Lab, Sharif University of Technology	Fall 2019 - Fall 2020
Teaching Assistant, Fundamentals of Programming with Python, Sharif University of Technology	Fall 2017
Olympiad Teacher, Astrophysics and Celestial Mechanics, Farzanegan 3 High School	Summer 2016